Mauna Loa CO2 Time Series Analysis

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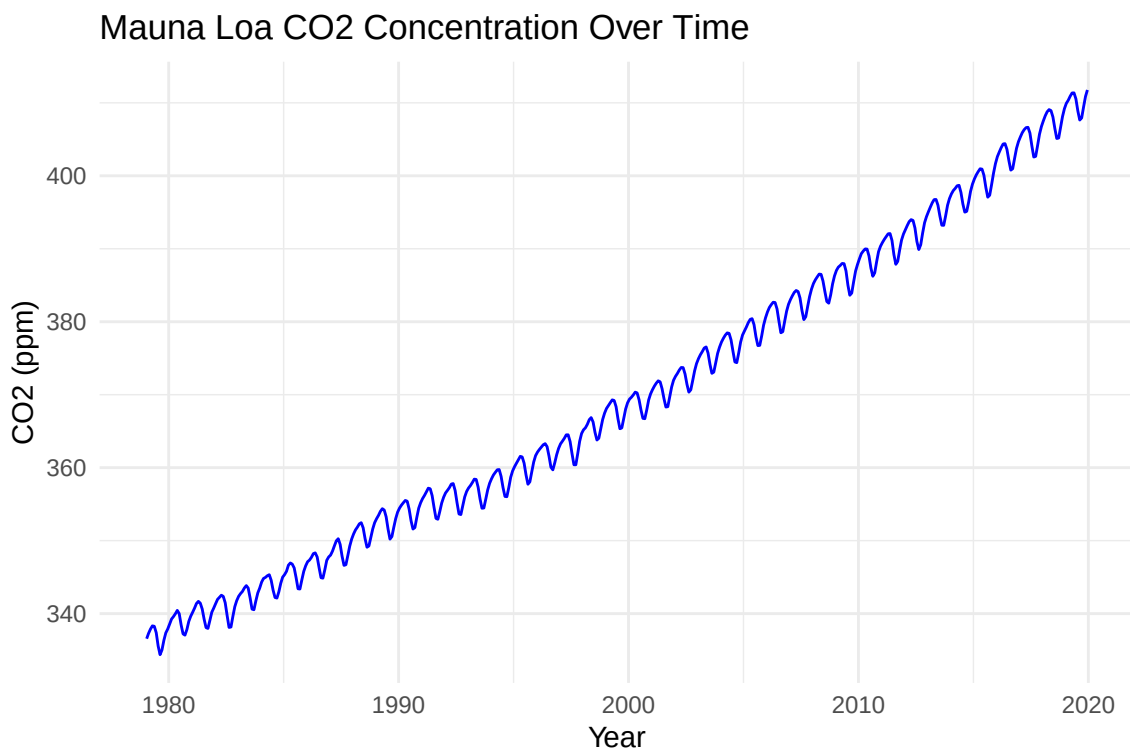
```r
# adding libraries
library(tidyverse)
library(fpp3)
library(ggplot2)
library(dplyr)
library(nortest)
library(tseries)
library(urca)



file <- "co2_mm_gl.csv"
climate_data <- read_csv(file, skip = 38)
head(climate_data)
```

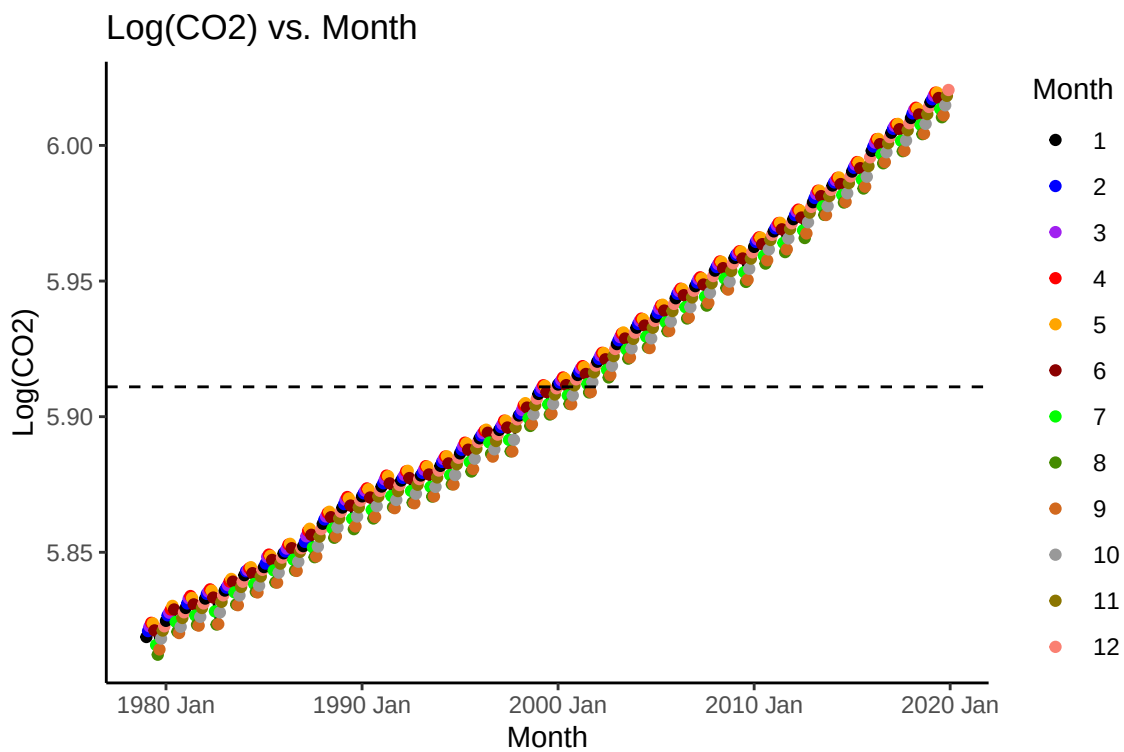
```
## # A tibble: 6 x 7
##    year month decimal average average_unc trend trend_unc
##   <dbl> <dbl>   <dbl>   <dbl>       <dbl> <dbl>     <dbl>
## 1  1979     1   1979.    337.        0.11  336.      0.1
## 2  1979     2   1979.    337.        0.09  336.      0.1
## 3  1979     3   1979.    338.        0.11  337.      0.1
## 4  1979     4   1979.    338.        0.13  337.      0.11
## 5  1979     5   1979.    338.        0.04  337.      0.11
## 6  1979     6   1979.    337.        0.18  337.      0.11
```

```r
# filter data to only before 2020
climate_data_clean <- climate_data %>%
    filter(year < 2020)
```

```r
# plot of CO2 over the years
ggplot(climate_data_clean, aes(x = decimal, y = average)) + geom_line(color = "blue") +
    labs(title = "Mauna Loa CO2 Concentration Over Time", x = "Year",
        y = "CO2 (ppm)") + theme_minimal()
```

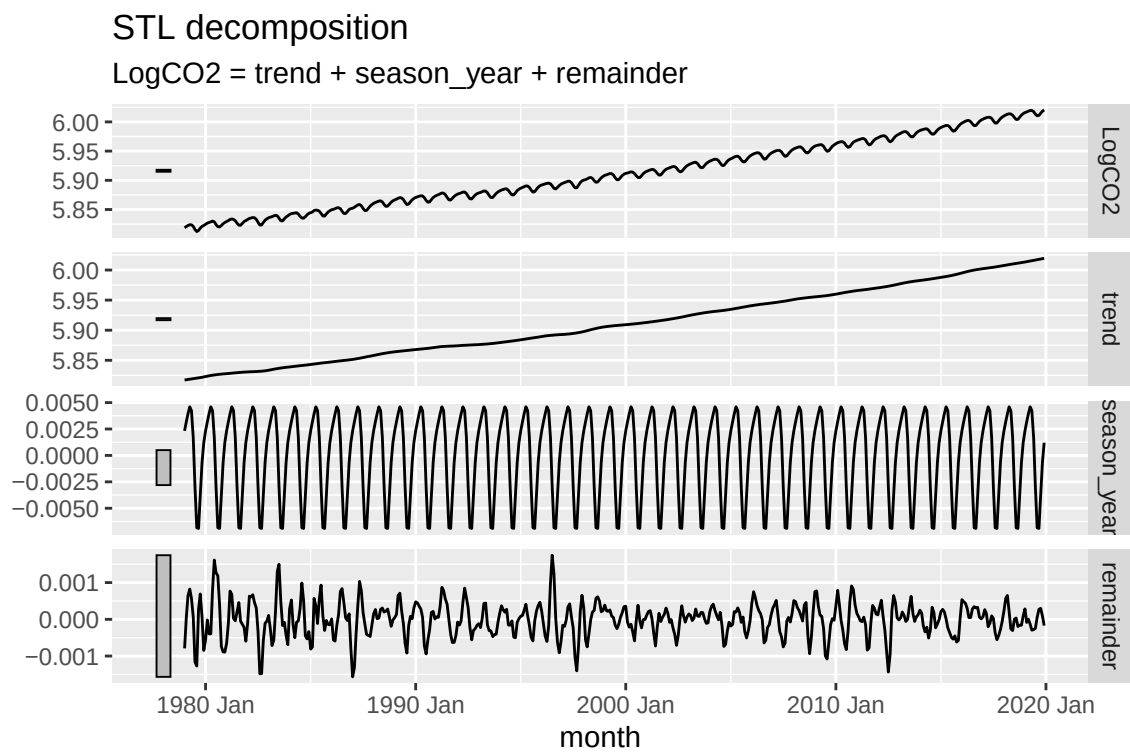



```
# dividing the data by month
climate_data_ts <- tsibble(month = yearmonth(seq(as.Date("1979-01-01"),
    as.Date("2019-12-01"), by = "1 month")), Time = climate_data_clean$decimal,
    CO2 = climate_data_clean$average, index = month)


Month <- as_factor(month(climate_data_ts$month))

climate_data_ts <- climate_data_ts %>%
    mutate(LogCO2 = log(CO2))

# plot of Log(CO2) over the years by month

mean_LogCO2 <- mean(climate_data_ts$LogCO2)
climate_data_ts %>%
    autoplot(LogCO2) + geom_point(aes(y = LogCO2, color = Month)) +
    scale_color_manual(values = c("black", "blue", "purple",
        "red", "orange", "darkred", "green", "chartreuse4", "chocolate",
        "gray60", "gold4", "salmon")) + geom_hline(aes(yintercept = mean_LogCO2),
    lty = 2) + ggtitle("Log(CO2) vs. Month") + xlab("Month") +
    ylab("Log(CO2)") + theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), panel.background = element_blank(),
    axis.line = element_line(colour = "black"))
```

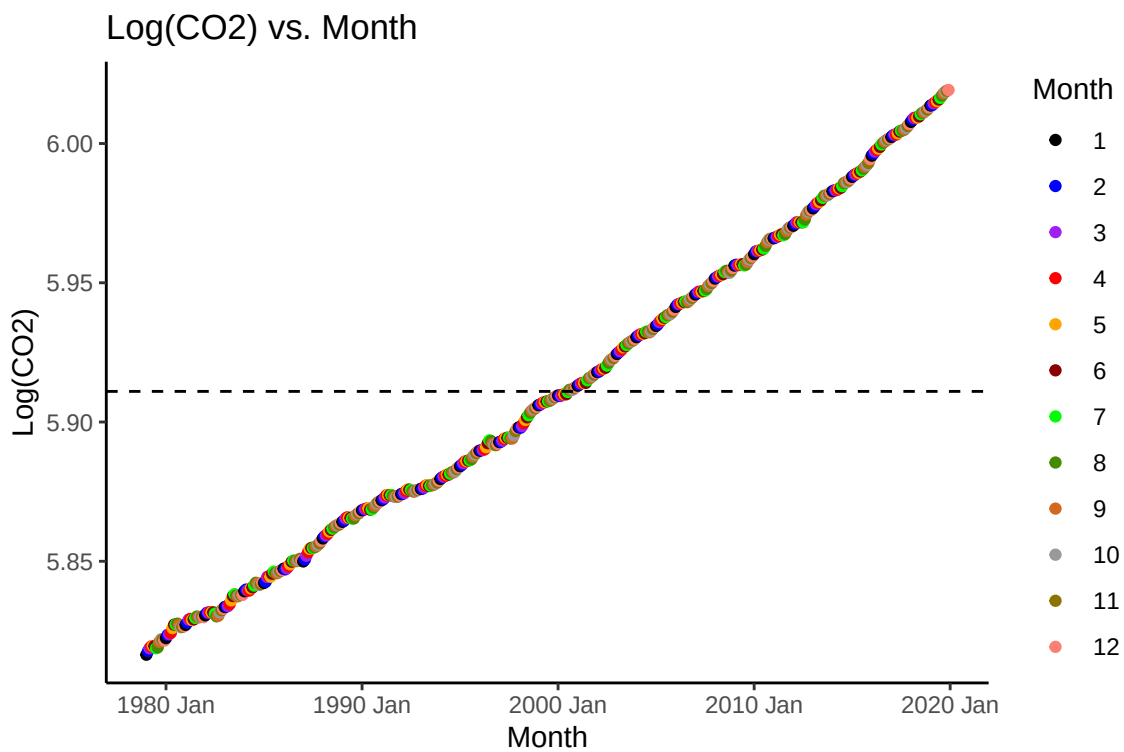


Visually the plot appears non-stationary but checking through the tests.

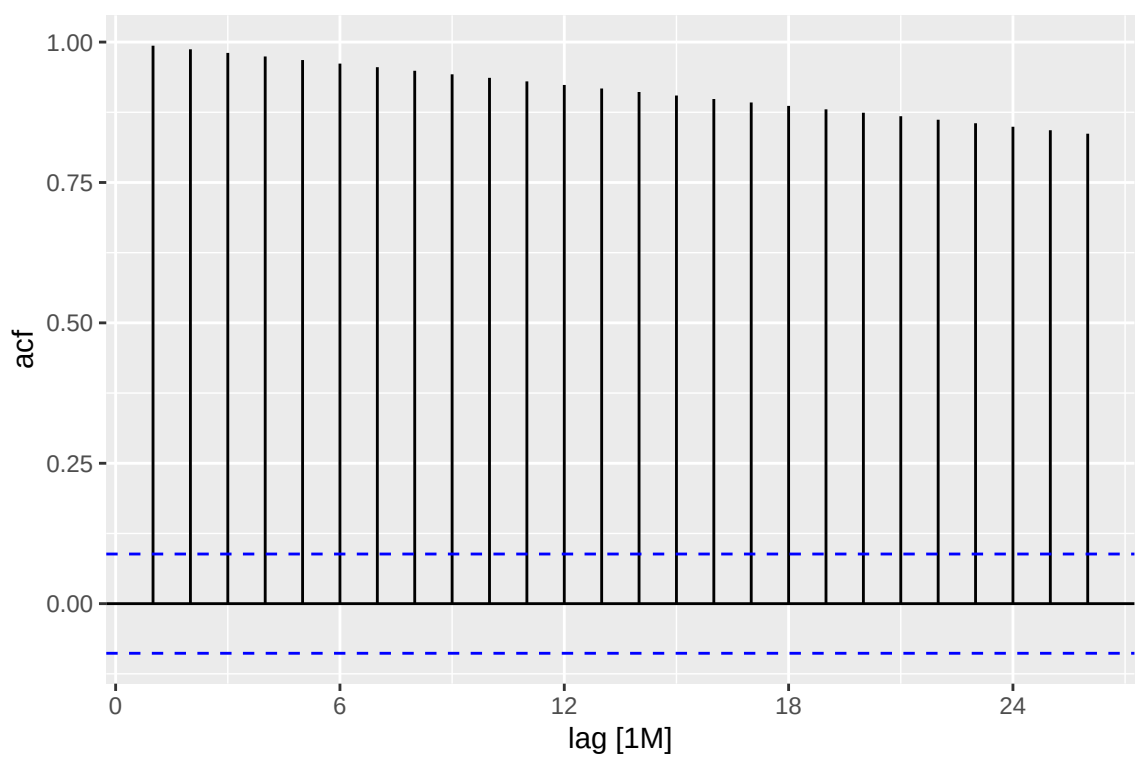
```
time_series_components <- climate_data_ts %>%
    model(STL(LogCO2 ~ season(window = "periodic"))) %>%
    components()

climate_data_ts <- climate_data_ts %>%
    add_column(seasonal = time_series_components$season_year,
        LogCO2_SA = time_series_components$season_adjust)

autoplot(time_series_components)
```




```
mean_LogCO2 <- mean(climate_data_ts$LogCO2_SA)
climate_data_ts %>%
    autoplot(LogCO2_SA) + geom_point(aes(y = LogCO2_SA, color = Month)) +
    scale_color_manual(values = c("black", "blue", "purple",
        "red", "orange", "darkred", "green", "chartreuse4", "chocolate",
        "gray60", "gold4", "salmon")) + geom_hline(aes(yintercept = mean_LogCO2),
    lty = 2) + ggtitle("Log(CO2) vs. Month") + xlab("Month") +
    ylab("Log(CO2)") + theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), panel.background = element_blank(),
    axis.line = element_line(colour = "black"))
```




```
climate_data_ts %>%
    ACF(LogCO2_SA) %>%
    autoplot()
```



```r
unitroot_kpss(climate_data_ts$LogCO2_SA)
```

```
##   kpss_stat kpss_pvalue
##    8.247788    0.010000
```

```r
adf.test(climate_data_ts$LogCO2_SA)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  climate_data_ts$LogCO2_SA
## Dickey-Fuller = -0.37114, Lag order = 7, p-value = 0.9877
## alternative hypothesis: stationary
```

The time series is clearly non-stationary because the p-value from the KPSS test is 0.01 which is less than 0.05. (not sure how to interpret Kpss test, need to add things about the ACF plot) And the large p-value of ADF test is evidence that the time series is non-stationary.

```r
# Computing first differences
climate_data_ts <- climate_data_ts %>%
    mutate(diff_LogCO2_SA = difference(LogCO2_SA))
head(climate_data_ts)
```

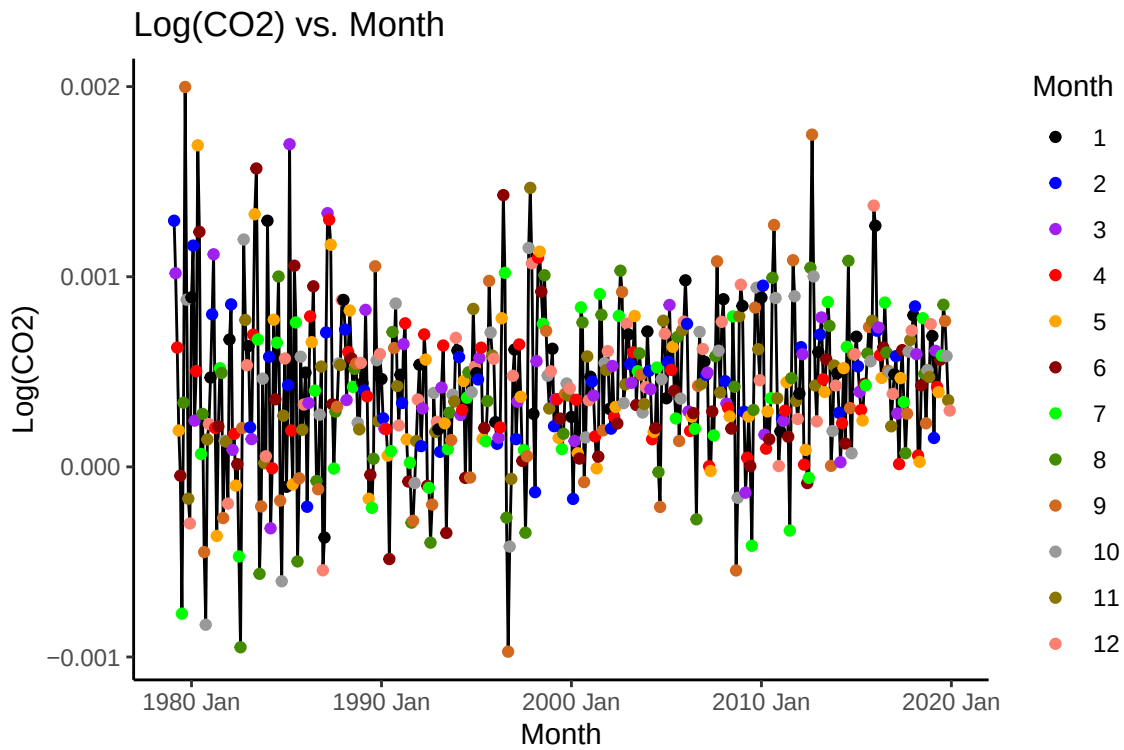
```
## # A tsibble: 6 x 7 [1M]
##      month  Time   CO2 LogCO2 seasonal LogCO2_SA diff_LogCO2_SA
##      <mth> <dbl> <dbl>  <dbl>    <dbl>     <dbl>          <dbl>
## 1 1979 Jan 1979.  337.   5.82  0.00233      5.82     NA
## 2 1979 Feb 1979.  337.   5.82  0.00320      5.82      0.00129
## 3 1979 Mar 1979.  338.   5.82  0.00393      5.82      0.00102
## 4 1979 Apr 1979.  338.   5.82  0.00460      5.82      0.000627
## 5 1979 May 1979.  338.   5.82  0.00423      5.82      0.000191
## 6 1979 Jun 1979.  337.   5.82  0.00168      5.82     -0.0000463
```

```r
tail(climate_data_ts)
```

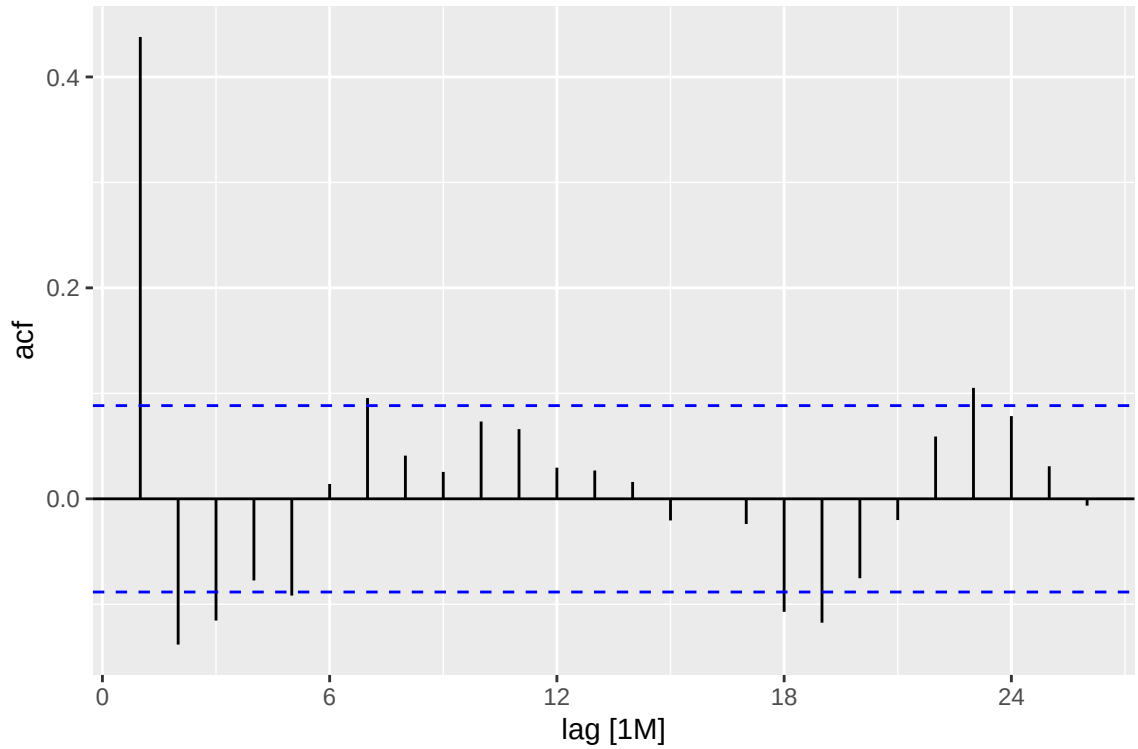
```
## # A tsibble: 6 x 7 [1M]
##      month  Time   CO2 LogCO2  seasonal LogCO2_SA diff_LogCO2_SA
##      <mth> <dbl> <dbl>  <dbl>     <dbl>     <dbl>          <dbl>
## 1 2019 Jul 2020.  409.   6.01 -0.00296       6.02       0.000585
## 2 2019 Aug 2020.  408.   6.01 -0.00685       6.02       0.000853
## 3 2019 Sep 2020.  408.   6.01 -0.00691       6.02       0.000767
## 4 2019 Oct 2020.  409.   6.01 -0.00380       6.02       0.000582
## 5 2019 Nov 2020.  411.   6.02 -0.000659      6.02       0.000351
## 6 2019 Dec 2020.  412.   6.02  0.00121       6.02       0.000296
```

```r
mean_diff_LogCO2_SA <- mean(climate_data_ts$diff_LogCO2_SA)

climate_data_ts %>%
    autoplot(diff_LogCO2_SA) + geom_point(aes(y = diff_LogCO2_SA,
    color = Month)) + scale_color_manual(values = c("black",
```

```
    "blue", "purple", "red", "orange", "darkred", "green", "chartreuse4",
    "chocolate", "gray60", "gold4", "salmon")) + geom_hline(aes(yintercept = mean_diff_LogCO2_SA),
    lty = 2) + ggtitle("Log(CO2) vs. Month") + xlab("Month") +
    ylab("Log(CO2)") + theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), panel.background = element_blank(),
    axis.line = element_line(colour = "black"))
```




```
climate_data_ts %>%
    ACF(diff_LogCO2_SA) %>%
    autoplot()
```




```r
unitroot_kpss(climate_data_ts$diff_LogCO2_SA)
```

```
##   kpss_stat kpss_pvalue
##  0.59597200  0.02300255
```

```r
adf.test(climate_data_ts$diff_LogCO2_SA[2:492])
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  climate_data_ts$diff_LogCO2_SA[2:492]
## Dickey-Fuller = -8.4187, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

(not sure how to interpret Kpss test, need to add things about the ACF plot) Since the p-value of the ADF test is less than 0.05, there is evidence to reject the null hypothesis in favor of the alternative hypothesis that the time series is stationary.

```r
# choosing ARIMA model

result_dcmp_ARIMA_SNAIVE <- climate_data_ts %>%
    model(decomposition_model(STL(CO2 ~ season(window = "periodic")),
        ARIMA(season_adjust ~ pdq(q = 0) + PDQ(0, 0, 0), stepwise = FALSE,
            approximation = FALSE, trace = TRUE), SNAIVE(season_year)))
```

```
## ARIMA(0,2,0)(0,0,0)[12]       -422.462803
## ARIMA(1,2,0)(0,0,0)[12]       -421.081477
```

```
## ARIMA(2,2,0)(0,0,0)[12]      -564.440275
## ARIMA(3,2,0)(0,0,0)[12]      -562.992145
## ARIMA(4,2,0)(0,0,0)[12]      -609.372819
## ARIMA(5,2,0)(0,0,0)[12]      -614.389848
```

```r
report(result_dcmp_ARIMA_SNAIVE)
```

```
## Series: CO2
## Model: STL decomposition model
## Combination: season_adjust + season_year
##
## ========================================
##
## Series: season_adjust
## Model: ARIMA(5,2,0)
##
## Coefficients:
##          ar1      ar2      ar3      ar4      ar5
##       0.0439  -0.6585  -0.0256  -0.2974  -0.1202
## s.e.  0.0450   0.0428   0.0524   0.0427   0.0450
##
## sigma^2 estimated as 0.0169:  log likelihood=313.28
## AIC=-614.56   AICc=-614.39   BIC=-589.4
##
## Series: season_year
## Model: SNAIVE
##
## sigma^2: 0
```